practice questions for AP practical

1. Create an android application to pass the data from one activity to another activity in the same application using intent (P3)
-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows

```
package com.example.activitydemo

import android.content.Intent
import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.platform.LocalContext
import androidx.compose.ui.unit.dp

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            MainScreen()
        }
    }
}

@Composable
fun MainScreen(){

    val context = LocalContext.current
    var name by remember { mutableStateOf("") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ){

        Text(
            text = "Main Activity",
            style = MaterialTheme.typography.headlineLarge
        )
```

```
        Spacer(modifier = Modifier.height(20.dp))

        OutlinedTextField(
            value = name,
            onValueChange = { name = it },
            label = { Text("Enter Your Name") }
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = {

            val intent = Intent(context, SecondActivity::class.java)
            intent.putExtra("username", name)
            context.startActivity(intent)

        }) {
            Text("Send Data")
        }
    }
}
```

2. Creating SecondActivity.kt (Steps)

Right click java → com.example.activitydemo

Click New → Kotlin Class/File

Name the file SecondActivity

Select Class and click OK

3. Modifying SecondActivity.kt

Edit the SecondActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp

class SecondActivity : ComponentActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        val receivedName = intent.getStringExtra("username")

        setContent {
            SecondScreen(receivedName ?: "")
        }
    }
}

@Composable
fun SecondScreen(name: String){

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ){

        Text(
            text = "Second Activity",
            style = MaterialTheme.typography.headlineLarge
        )

        Spacer(modifier = Modifier.height(20.dp))

        Text(
            text = "Received Data: $name",
            style = MaterialTheme.typography.bodyLarge
        )
    }
}
```

4. Modifying AndroidManifest.xml

Add SecondActivity inside the <application> tag.

```
<activity android:name=".SecondActivity"/>
```

5. Configuring build.gradle.kts (Module:app)

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
```

```
    namespace = "com.example.activitydemo"
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
}
```

7. Configuring gradle-wrapper.properties

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
```

zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
---

2. Create a user interface with a TextView, Button, and ImageView. When the button is clicked, the image should be changed dynamically
-->
1. Modifying MainActivity.kt

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.Image
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.res.painterResource
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            MainScreen()
        }
    }
}

@Composable
fun MainScreen(){

    var imageChanged by remember { mutableStateOf(false) }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally
    ){

        Text(
            text = "Dynamic Image Change",
            style = MaterialTheme.typography.headlineLarge
        )
```

```
        Spacer(modifier = Modifier.height(20.dp))

        Image(
            painter = painterResource(
                id = if(imageChanged)
                    R.drawable.image2
                else
                    R.drawable.image1
            ),
            contentDescription = "Demo Image",
            modifier = Modifier.size(200.dp)
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = {
            imageChanged = !imageChanged
        }) {
            Text("Change Image")
        }
    }
}
```

2. Adding Images to Drawable (Steps)

Open res → drawable folder.

Add two images.

Example:
image1.png
image2.png

These images will be used to dynamically change the ImageView when the button is clicked.

3. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
    compileSdk = 35
```

```
    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.8.0")
}
```

4. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
---

3. Implement a custom color scheme for your registration form. Set the background color of the entire layout to a light color, and use a contrasting color for the Submit button.
-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.graphics.Color
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            RegistrationForm()
        }
    }
}

@Composable
fun RegistrationForm(){

    Column(
        modifier = Modifier
            .fillMaxSize()
            .background(Color(0xFFEAF6FF))   // Light background color
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally
```

```
    ){

        Text(
            text = "Registration Form",
            style = MaterialTheme.typography.headlineLarge
        )

        Spacer(modifier = Modifier.height(20.dp))

        OutlinedTextField(
            value = "",
            onValueChange = {},
            label = { Text("Enter Name") }
        )

        Spacer(modifier = Modifier.height(10.dp))

        OutlinedTextField(
            value = "",
            onValueChange = {},
            label = { Text("Enter Email") }
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(
            onClick = { },
            colors = ButtonDefaults.buttonColors(
                containerColor = Color(0xFF1976D2)   // Contrasting button color
            )
        ) {
            Text("Submit", color = Color.White)
        }
    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
    compileSdk = 35
```

```
    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.8.0")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

4. Insert the new contents in the following resources and demonstrate their uses in the android application
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)(P2)

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.Image
import androidx.compose.foundation.background
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.res.colorResource
import androidx.compose.ui.res.dimensionResource
import androidx.compose.ui.res.painterResource
import androidx.compose.ui.res.stringResource
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            ResourceDemo()
        }
    }
}

@Composable
fun ResourceDemo(){

    Column(
        modifier = Modifier
```

```
            .fillMaxSize()
            .background(colorResource(id = R.color.backgroundColor))
            .padding(dimensionResource(id = R.dimen.padding_large)),
        horizontalAlignment = Alignment.CenterHorizontally
    ){

        Text(
            text = stringResource(id = R.string.app_title),
            style = MaterialTheme.typography.headlineLarge,
            color = colorResource(id = R.color.textColor)
        )

        Spacer(modifier = Modifier.height(20.dp))

        Image(
            painter = painterResource(id = R.drawable.sample_image),
            contentDescription = "Sample Image",
            modifier = Modifier.size(200.dp)
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = { }) {
            Text(stringResource(id = R.string.button_text))
        }
    }
}
```

2. Modifying strings.xml

Insert the following contents in res/values/strings.xml:

```
<resources>
    <string name="app_name">ResourceDemoApp</string>
    <string name="app_title">Android Resource Demo</string>
    <string name="button_text">Click Me</string>
</resources>
```

3. Modifying colors.xml

Insert the following contents in res/values/colors.xml:

```
<resources>
    <color name="backgroundColor">#E3F2FD</color>
    <color name="textColor">#0D47A1</color>
</resources>
```

4. Modifying dimens.xml

Create res/values/dimens.xml and insert:

```
<resources>
   <dimen name="padding_large">24dp</dimen>
</resources>
```

5. Adding Drawable Resource

Add an image in res/drawable folder.

Example:

sample_image.png

This image will be used in the Image component.

6. Modifying themes.xml

Insert or modify res/values/themes.xml:

```
<resources xmlns:tools="http://schemas.android.com/tools">
   <style name="Theme.ActivityDemo" parent="Theme.Material3.DayNight.NoActionBar">
      <item name="colorPrimary">@color/textColor</item>
   </style>
</resources>
```

7. Configuring build.gradle.kts (Module:app)

```
plugins {
   id("com.android.application")
   id("org.jetbrains.kotlin.android")
}

android {
   namespace = "com.example.activitydemo"
   compileSdk = 35

   defaultConfig {
      applicationId = "com.example.activitydemo"
      minSdk = 24
      targetSdk = 35
      versionCode = 1
      versionName = "1.0"
   }

   buildFeatures {
      compose = true
```

```
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
}
```

8. Configuring gradle-wrapper.properties

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

5. Create an android application and print “hello world” program(P1)
-->
1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
```

```
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.Composable
import androidx.compose.ui.Modifier
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            HelloWorldScreen()
        }
    }
}

@Composable
fun HelloWorldScreen(){

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp),
        horizontalAlignment = Alignment.CenterHorizontally,
        verticalArrangement = Arrangement.Center
    ){

        Text(
            text = "Hello World",
            style = MaterialTheme.typography.headlineLarge
        )

    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
```

```
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

6. Create a simple Registration Form with the following fields:

- Name (Text Input)
- Email (Text Input)
- Phone Number (Text Input)
- Password (Password Input)
- Confirm Password (Password Input)
- Submit Button

-->

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.PasswordVisualTransformation
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
   override fun onCreate(savedInstanceState: Bundle?) {
      super.onCreate(savedInstanceState)

      setContent {
         RegistrationForm()
      }
   }
}

@Composable
fun RegistrationForm(){

   var name by remember { mutableStateOf("") }
   var email by remember { mutableStateOf("") }
```

```
var phone by remember { mutableStateOf("") }
var password by remember { mutableStateOf("") }
var confirmPassword by remember { mutableStateOf("") }

Column(
    modifier = Modifier
        .fillMaxSize()
        .padding(16.dp),
    horizontalAlignment = Alignment.CenterHorizontally
){

    Text(
        text = "Registration Form",
        style = MaterialTheme.typography.headlineLarge
    )

    Spacer(modifier = Modifier.height(20.dp))

    OutlinedTextField(
        value = name,
        onValueChange = { name = it },
        label = { Text("Name") }
    )

    Spacer(modifier = Modifier.height(10.dp))

    OutlinedTextField(
        value = email,
        onValueChange = { email = it },
        label = { Text("Email") }
    )

    Spacer(modifier = Modifier.height(10.dp))

    OutlinedTextField(
        value = phone,
        onValueChange = { phone = it },
        label = { Text("Phone Number") }
    )

    Spacer(modifier = Modifier.height(10.dp))

    OutlinedTextField(
        value = password,
        onValueChange = { password = it },
        label = { Text("Password") },
        visualTransformation = PasswordVisualTransformation()
    )
```

```
        Spacer(modifier = Modifier.height(10.dp))

        OutlinedTextField(
            value = confirmPassword,
            onValueChange = { confirmPassword = it },
            label = { Text("Confirm Password") },
            visualTransformation = PasswordVisualTransformation()
        )

        Spacer(modifier = Modifier.height(20.dp))

        Button(onClick = { }) {
            Text("Submit")
        }
    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
```

```
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.8.0")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```

---

7. Create a Registration Form with the following fields:

- Username (Text Input)
- Email Address (Text Input)
- Gender (Radio Buttons with options "Male" and "Female")
- Password (Password Input)
- Submit Button

-->

1. Modifying MainActivity.kt

Edit the MainActivity.kt file as follows:

```
package com.example.activitydemo

import android.os.Bundle
import androidx.activity.ComponentActivity
import androidx.activity.compose.setContent
import androidx.compose.foundation.layout.*
import androidx.compose.material3.*
import androidx.compose.runtime.*
import androidx.compose.ui.Modifier
import androidx.compose.ui.text.input.PasswordVisualTransformation
import androidx.compose.ui.unit.dp
import androidx.compose.ui.Alignment

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)

        setContent {
            RegistrationForm()
        }
    }
}

@Composable
fun RegistrationForm(){

    var username by remember { mutableStateOf("") }
    var email by remember { mutableStateOf("") }
    var password by remember { mutableStateOf("") }
    var gender by remember { mutableStateOf("Male") }

    Column(
        modifier = Modifier
            .fillMaxSize()
            .padding(16.dp)
    ){

        Text(
            text = "Registration Form",
            style = MaterialTheme.typography.headlineLarge
        )

        Spacer(modifier = Modifier.height(20.dp))

        OutlinedTextField(
            value = username,
```

```
    onValueChange = { username = it },
    label = { Text("Username") }
)

Spacer(modifier = Modifier.height(10.dp))

OutlinedTextField(
    value = email,
    onValueChange = { email = it },
    label = { Text("Email Address") }
)

Spacer(modifier = Modifier.height(20.dp))

Text("Select Gender")

Row(
    verticalAlignment = Alignment.CenterVertically
){

    RadioButton(
        selected = gender == "Male",
        onClick = { gender = "Male" }
    )
    Text("Male")

    Spacer(modifier = Modifier.width(20.dp))

    RadioButton(
        selected = gender == "Female",
        onClick = { gender = "Female" }
    )
    Text("Female")
}

Spacer(modifier = Modifier.height(20.dp))

OutlinedTextField(
    value = password,
    onValueChange = { password = it },
    label = { Text("Password") },
    visualTransformation = PasswordVisualTransformation()
)

Spacer(modifier = Modifier.height(20.dp))

Button(onClick = { }) {
    Text("Submit")
```

```
        }
    }
}
```

2. Configuring build.gradle.kts (Module:app)

Modify build.gradle.kts (Module: app) as follows:

```
plugins {
    id("com.android.application")
    id("org.jetbrains.kotlin.android")
}

android {
    namespace = "com.example.activitydemo"
    compileSdk = 35

    defaultConfig {
        applicationId = "com.example.activitydemo"
        minSdk = 24
        targetSdk = 35
        versionCode = 1
        versionName = "1.0"

        testInstrumentationRunner =
            "androidx.test.runner.AndroidJUnitRunner"
    }

    buildFeatures {
        compose = true
    }

    composeOptions {
        kotlinCompilerExtensionVersion = "1.4.3"
    }

    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_1_8
        targetCompatibility = JavaVersion.VERSION_1_8
    }

    kotlinOptions {
        jvmTarget = "1.8"
    }
}

dependencies {

    implementation("androidx.core:core-ktx:1.9.0")
```

```
    implementation("androidx.lifecycle:lifecycle-runtime-ktx:2.7.0")
    implementation("androidx.activity:activity-compose:1.10.0")

    implementation(platform("androidx.compose:compose-bom:2023.03.00"))
    implementation("androidx.compose.ui:ui")
    implementation("androidx.compose.ui:ui-graphics")
    implementation("androidx.compose.ui:ui-tooling-preview")
    implementation("androidx.compose.material3:material3")

    implementation("androidx.appcompat:appcompat:1.6.1")
    implementation("com.google.android.material:material:1.8.0")
}
```

3. Configuring gradle-wrapper.properties

Set the gradle-wrapper.properties correctly:

```
distributionBase=GRADLE_USER_HOME
distributionPath=wrapper/dists
distributionUrl=https\://services.gradle.org/distributions/gradle-8.5-bin.zip
zipStoreBase=GRADLE_USER_HOME
zipStorePath=wrapper/dists
```